Model Predictive Control

Summary and Outlook

Prof. Melanie Zeilinger

ETH Zurich

Spring 2026

Goal: Optimal Constrained Control

$$x(k+1) = g(x(k), u(k)) \qquad\qquad x, u \in \mathcal{X}, \mathcal{U}$$

Design control law $u(k) = \kappa(x(k))$ such that the system:

1. Satifies constraints : $\{x(k)\} \subset \mathcal{X}$, $\{u(k)\} \subset \mathcal{U}$
2. Is stable: $\lim_{k\to\infty} x(k) = 0$
3. Optimizes "performance"
4. Maximizes the set $\{x(0) \,|\, \text{Conditions 1-3 are met}\}$

What we have learned so far

Chapter 1	System Theory Basics
Chapter 2	Optimal Control of Unconstrained Systems
Chapter 3	Basics on Optimization
Chapter 4	Constrained Finite Time Optimal Control
Chapter 5	Introduction to Constrained Systems
Chapter 6	MPC Feasibility and Stability

What we have learned so far

MPC design process:

1. Obtain a model of the system
2. (Design a state observer)
3. Define finite time optimal control problem
4. Set up optimization problem
5. Solve optimization problem to get optimal input
6. Verify the closed-loop system

What we have learned so far

MPC design process:

1.	Obtain a model of the system	
	$\rightarrow$ *Discretization, linearization*	Chapter 1
2.	(Design a state observer)	
3.	Define finite time optimal control problem	Chapter 2,4
	$\rightarrow$ Define *constraints* (convex sets)	Chapter 3
4.	Set up optimization problem	
	$\rightarrow$ QP, LP formulation	Chapter 4
5.	Solve optimization problem to get optimal input	
	$\rightarrow$ Convex problems, optimality conditions	Chapter 3
6.	Verify the closed-loop system	
	$\rightarrow$ Choose *terminal cost and constraint*	Chapter 6
	using *invariance* and *stability*	Chapter 1,5
	e.g. based on *infinite horizon LQR* controller	Chapter 2

What's next?

09.04.	*Easter break*
Lecture 8 16.04.	MPC Practical Considerations & Tracking and Soft-Constrained MPC
Lecture 9 23.04.	Robust MPC I
Lecture 10 30.04.	Robust MPC II
Lecture 11 07.05.	Implementation Methods
14.05.	*Ascension Day*
Lecture 12	Advanced Topics
21.05.	
Lecture 13 28.05.	Invited Presentations on MPC Applications